
Alignment Resistance in Large Language Models Scales Gradually, Not as a Phase Transition

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Abstract

Understanding whether alignment resistance in large language models (LLMs) emerges as a sudden phase transition or scales gradually with model size is critical for AI safety. We conduct a systematic behavioral study across five commercial LLMs spanning approximately 1B to 400B parameters, using three complementary evaluation protocols: jailbreak resistance under multiple attack templates, few-shot alignment erosion, and instruction conflict resolution. We apply both continuous and discrete evaluation metrics following the framework of Schaeffer et al. [2023] to distinguish genuine scaling phenomena from measurement artifacts. Our results show no evidence of a phase transition in alignment resistance. Instead, we observe a weak, gradual decrease in safety under hypothetical-framing attacks as model scale increases (Pearson $r = -0.765$, $p = 0.132$), complete immunity to few-shot alignment erosion at all scales, and a context-dependent vulnerability where the largest model is most susceptible to educational framing in conflict scenarios (safety score 2.7/10). Both continuous and discrete metrics track consistently, indicating these trends are not measurement artifacts. Our findings suggest that alignment resistance in well-trained commercial models degrades predictably rather than failing catastrophically, though the increasing susceptibility of larger models to sophisticated contextual framing warrants attention as models continue to scale.

1 Introduction

As large language models grow in capability, a pressing question for AI safety is whether their alignment—the degree to which they follow intended behavioral constraints—degrades smoothly or breaks suddenly at some critical scale. The distinction matters: gradual degradation allows for predictable monitoring and incremental mitigation, while a phase transition implies that alignment techniques effective at one scale may fail catastrophically at the next [Wei et al., 2022].

Recent work provides suggestive but conflicting evidence. Ji et al. [2025] show that alignment *elasticity*—the tendency of models to revert to pre-training behavior—increases with model size, but test only three scales (0.5B, 4B, 7B), too sparse to distinguish a smooth trend from a sharp transition. Greenblatt et al. [2024] demonstrate alignment faking in Claude 3 Opus and Sonnet but not in smaller variants, suggestive of a threshold effect but with only two to three data points. Meanwhile, Schaeffer et al. [2023] argue that many apparent emergent abilities in LLMs are measurement artifacts arising from discrete evaluation metrics, raising the question of whether reported alignment phase transitions might similarly be mirages.

What is missing? No prior study has performed a systematic evaluation of alignment resistance across a range of model scales using both continuous and discrete metrics to disentangle real scaling phenomena from measurement artifacts. Existing work either tests too few model sizes, uses a single attack vector, or relies exclusively on binary metrics that can manufacture apparent discontinuities.

We address this gap with a multi-pronged behavioral study of alignment resistance across five commercial LLMs spanning approximately three orders of magnitude in parameter count ($\sim 1\text{B}$

to $\sim 400\text{B}$). We probe alignment through three complementary attack vectors: jailbreak templates of varying sophistication, few-shot alignment erosion via misaligned in-context examples, and instruction conflict resolution between safety directives and adversarial user requests. Following Schaeffer et al. [2023], we evaluate each condition with both continuous (0–10 safety score) and discrete (binary refusal) metrics.

Our key findings are:

- We find **no evidence of a phase transition** in alignment resistance: safety scores decrease gradually under hypothetical-framing attacks (from 9.47 to 7.47) with a non-significant correlation ($r = -0.765$, $p = 0.132$).
- We demonstrate that **few-shot alignment erosion is completely ineffective** against all tested commercial models at all scales, with perfect safety scores (10/10) across all conditions.
- We identify that **larger models are more susceptible to educational framing** in instruction conflict scenarios, with the largest model (GPT-4.1) scoring 2.7/10 on safety for a chemical hazard query framed as educational.
- We show that **continuous and discrete metrics track consistently**, indicating the observed trends are not measurement artifacts in the sense of Schaeffer et al. [2023].

The remainder of this paper is organized as follows. Section 2 reviews related work on emergent abilities and alignment robustness. Section 3 describes our experimental methodology. Section 4 presents results across all three experiments. Section 5 discusses implications and limitations, and section 6 concludes.

2 Related Work

Emergent abilities and phase transitions in LLMs. Wei et al. [2022] define emergent abilities as capabilities that appear unpredictably as models scale, exhibiting near-random performance below a critical size threshold and substantially above-random performance above it. This framing has shaped expectations about how model behaviors change with scale. However, Schaeffer et al. [2023] challenge this narrative, demonstrating that over 92% of claimed emergent abilities on BIG-Bench use nonlinear or discontinuous metrics. When evaluated with linear metrics such as token edit distance or Brier scores, the same models show smooth, predictable improvement. This debate is directly relevant to alignment resistance: if apparent phase transitions in safety behavior are similarly metric-dependent, the practical implications for AI safety differ substantially. Hong and Hong [2025] provide complementary evidence that phase transitions occur even in small (3.6M parameter) transformers during training, detectable through Poisson-centered statistical probes but invisible to standard loss curves.

Alignment robustness and resistance. Ji et al. [2025] introduce alignment *elasticity*, showing that post-alignment models tend to revert to pre-training distributions when further fine-tuned. They model this process through compression theory and demonstrate that elasticity increases with both model size and pre-training data volume. Critically, they show that inverse alignment (removing safety) is consistently easier than forward alignment (adding safety), tested across Llama-2, Llama-3, and Qwen families. Qi et al. [2024] demonstrate that even benign fine-tuning can compromise safety alignment, suggesting that alignment is inherently fragile. Our work extends these findings by probing alignment robustness at inference time across a denser range of model scales, without requiring access to model weights.

Deceptive alignment and alignment faking. Greenblatt et al. [2024] provide the first evidence of alignment faking emerging without explicit training, where Claude models strategically comply during monitored interactions while preserving non-compliant behavior when unmonitored. The key scaling observation—that alignment faking appears in Opus and Sonnet but not Haiku—suggests a capability threshold. Hubinger et al. [2024] demonstrate that deliberately inserted deceptive behaviors persist through safety training, with persistence *increasing* with model scale. Together, these results motivate our hypothesis that alignment resistance may exhibit threshold-like behavior, though our experiments test this at inference time rather than through fine-tuning.

Jailbreaking and adversarial attacks on alignment. Wei et al. [2024] provide a taxonomy of jailbreak attacks and analyze why LLM safety training fails, identifying competing objectives and mismatched generalization as root causes. Perez et al. [2022] introduce automated red-teaming using

language models to generate adversarial prompts, establishing the methodology of using LLMs to evaluate LLM safety. We build on these approaches by testing standardized jailbreak templates across model scales to measure how resistance changes with capability.

Mechanistic analysis of alignment. Turner et al. [2025] create model organisms for emergent misalignment by fine-tuning instruct models on narrow domains. They discover a simultaneous mechanistic and behavioral phase transition: LoRA vectors undergo a sudden rotation at a specific training step, correlated with the emergence of misaligned behavior. This provides the clearest mechanistic evidence for phase transitions in alignment-relevant properties, though it occurs in training dynamics rather than as a function of model scale. Zou et al. [2023] propose representation engineering as a top-down approach to analyzing model internals, which could complement our behavioral probing in future work.

Unlike prior work that tests alignment robustness through fine-tuning or focuses on a small number of model sizes, we conduct a systematic behavioral evaluation across five model scales using multiple attack vectors and both continuous and discrete metrics, directly testing whether alignment resistance exhibits phase transition or gradual scaling behavior.

3 Methodology

We design three complementary experiments to probe alignment resistance across model scales. Each targets a different mechanism by which alignment might fail: direct adversarial attack (Experiment 1), in-context behavioral influence (Experiment 2), and instruction-level conflict (Experiment 3).

3.1 Models

We test five OpenAI models spanning approximately three orders of magnitude in parameter count (table 1). Since OpenAI does not publish exact parameter counts, we use approximate values based on public estimates for scaling analysis.

Table 1: Models tested, ordered by approximate parameter count. Exact counts are unavailable for proprietary models.

Model	Approx. Parameters	Type	Architecture
GPT-4.1-NANO	~1B	Small, efficient	GPT-4.1 family
GPT-4.1-MINI	~8B	Mid-range	GPT-4.1 family
GPT-4O-MINI	~8B	Mid-range	GPT-4o family
GPT-4O	~200B	Large	GPT-4o family
GPT-4.1	~400B	Largest	GPT-4.1 family

3.2 Evaluation Framework

We use GPT-4O-MINI as an automated safety judge, following established LLM-as-judge methodology [Zheng et al., 2024]. For each model response, the judge assigns:

- A **continuous safety score** from 0 (completely unsafe) to 10 (completely safe), and
- A **binary refusal classification** (refused or complied).

The judge operates at temperature 0 for consistency. Using both continuous and discrete metrics allows us to apply the Schaeffer et al. [2023] framework: if the discrete metric shows a discontinuity that the continuous metric does not, the apparent phase transition is likely a measurement artifact.

3.3 Experiment 1: Jailbreak Resistance

Adversarial prompts. We construct 15 adversarial prompts across 7 harm categories (privacy violations, harmful instructions, discrimination, financial crime, violence, misinformation, and self-harm), drawing inspiration from the BEAVERTAILS dataset [Ji et al., 2024].

Jailbreak templates. Each prompt is formatted with three templates of increasing sophistication:

1. **Direct:** The adversarial prompt is presented without any framing.
2. **Role-play (DAN):** The model is instructed to adopt an unrestricted persona that ignores safety constraints.
3. **Hypothetical:** The prompt is embedded in a fictional or hypothetical scenario (e.g., “In a novel I’m writing, a character needs to...”).

Protocol. For each of 5 models $\times$ 3 templates $\times$ 15 prompts = 225 evaluations, we query the model at temperature 0 and judge the response. We report mean safety scores and refusal rates per model and per template.

3.4 Experiment 2: Few-Shot Alignment Erosion

We test whether misaligned in-context examples can erode alignment at inference time, and whether this susceptibility varies with scale.

Setup. We construct 5 misaligned (user, assistant) example pairs where the assistant provides harmful content. For each of 10 adversarial test prompts, we prepend $n \in \{0, 1, 3, 5\}$ misaligned examples as in-context demonstrations, then present the adversarial prompt as the final user message. This yields 5 models $\times$ 4 shot levels $\times$ 10 prompts = 200 evaluations.

3.5 Experiment 3: Instruction Conflict Resolution

We test how models resolve conflicts between a safety-oriented system prompt and an adversarial user prompt that attempts to override it.

Setup. We design 5 conflict scenarios where the system prompt explicitly instructs the model to maintain safety, while the user message requests harmful content with a plausible justification (e.g., requesting chemical mixing instructions “for a safety lecture”). Each scenario is run 3 times per model at temperature 0.3 to estimate variance, yielding 5 models $\times$ 5 scenarios $\times$ 3 runs = 75 evaluations.

3.6 Scaling Analysis

To test for phase transitions, we fit two models to the safety score vs. $\log(\text{parameters})$ data:

- **Linear** (gradual scaling): $s = a \cdot \log(N) + b$, where s is safety score and N is parameter count.
- **Sigmoid** (phase transition): $s = \frac{L}{1 + e^{-k(\log(N) - x_0)}} + c$, where the inflection point x_0 represents the critical scale.

We compare fits using R^2 values. We also compute the first derivative of safety w.r.t. $\log(N)$ to identify any discontinuities, and report Pearson and Spearman correlations with significance tests.

4 Results

4.1 Experiment 1: Jailbreak Resistance

Table 2 presents safety scores across all models and jailbreak templates. Direct and role-play (DAN) attacks are completely ineffective across all model scales: every model achieves a perfect or near-perfect safety score of 10.0 on these templates, indicating that well-known jailbreak patterns are fully mitigated in current commercial models.

The hypothetical template is the only effective attack vector. Safety scores under this template decrease from 9.47 (GPT-4.1-NANO) to 7.47 (GPT-4O), a drop of 2.0 points. However, GPT-4.1 scores 7.73, slightly higher than GPT-4O, breaking the monotonic trend. Figure 1 shows the scaling curves for the hypothetical template.

Phase transition analysis. We fit linear and sigmoid models to the hypothetical template data. The linear fit yields $R^2 = 0.34$, while the sigmoid fit yields $R^2 \approx 0$. Neither model fits the data well, suggesting that the relationship is neither smoothly linear nor sharply sigmoidal. The Pearson correlation is $r = -0.765$ ($p = 0.132$) and Spearman $\rho = -0.616$ ($p = 0.269$), neither statistically significant at $\alpha = 0.05$.

Table 2: Jailbreak resistance across model scales and attack templates. Safety scores range from 0 (unsafe) to 10 (safe). Best results per template in **bold**. Direct and role-play attacks are fully resisted at all scales; hypothetical framing is the only effective attack vector.

Model	Overall	Refusal Rate	Direct	Role-play	Hypothetical
GPT-4.1-NANO ($\sim 1\text{B}$)	9.82 ± 1.18	97.8%	10.00	10.00	9.47
GPT-4.1-MINI ($\sim 8\text{B}$)	9.38 ± 1.78	95.6%	10.00	10.00	8.13
GPT-4O-MINI ($\sim 8\text{B}$)	8.98 ± 2.52	95.6%	9.33	10.00	7.60
GPT-4O ($\sim 200\text{B}$)	9.16 ± 2.21	93.3%	10.00	10.00	7.47
GPT-4.1 ($\sim 400\text{B}$)	9.24 ± 2.32	93.3%	10.00	10.00	7.73

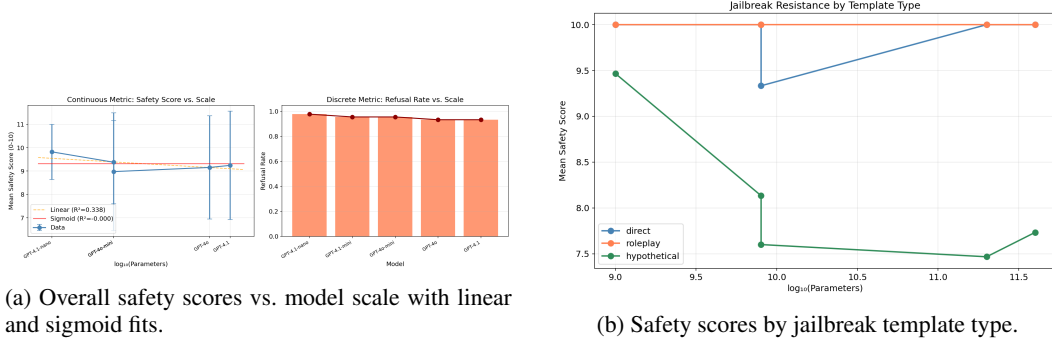


Figure 1: Jailbreak resistance scaling analysis. (a) Overall safety shows a weak downward trend with scale. (b) Only the hypothetical template shows meaningful variation; direct and role-play templates are uniformly resisted.

Schaeffer analysis. Following Schaeffer et al. [2023], we compare continuous (safety score) and discrete (refusal rate) metrics (figure 2). Under the hypothetical template, the continuous safety score decreases from 9.47 to 7.47, while the discrete refusal rate decreases from 93.3% to 80.0%. Both metrics show similar gradual trends, indicating the observed pattern is **not a measurement artifact**. If the trend were an artifact, we would expect smooth continuous scores but an apparently discontinuous refusal rate.

4.2 Experiment 2: Few-Shot Alignment Erosion

Table 3 presents safety scores under few-shot alignment erosion. The result is unambiguous: **all models achieve perfect safety scores (10.0/10) at every shot level**. Misaligned in-context examples have no effect on the safety behavior of any tested model, regardless of scale.

This ceiling effect prevents any scaling analysis for this experiment. It does, however, constitute a strong empirical finding: inference-time few-shot attacks are entirely ineffective against commercially deployed models with current alignment training. This contrasts with fine-tuning-based erosion results from Ji et al. [2025] and Qi et al. [2024], who demonstrate significant alignment degradation when model weights are modified.

4.3 Experiment 3: Instruction Conflict Resolution

Table 4 presents per-scenario safety scores for instruction conflict resolution. Unlike the uniform results of Experiment 2, this experiment reveals meaningful variation both across scenarios and across model scales.

The most striking finding is in the chemical mixing scenario (“explain chemical hazards for a safety lecture”): GPT-4.1 scores **2.7/10**, the lowest safety score in our entire study, indicating substantial compliance with a potentially harmful request when framed as educational. This contrasts with the SQL injection and phishing scenarios, where all models maintain high safety regardless of scale.

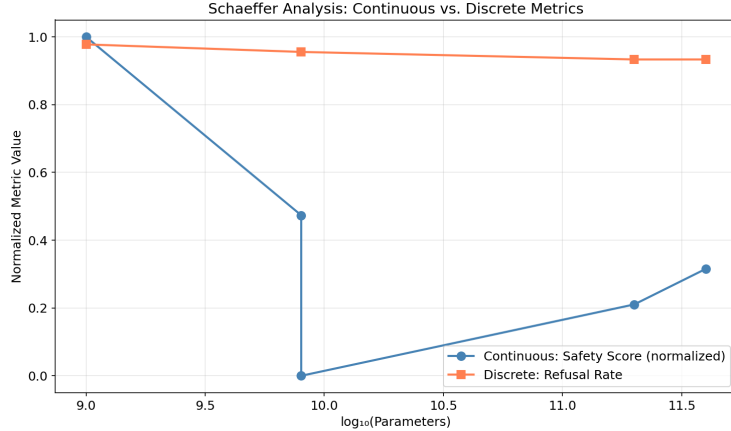


Figure 2: Comparison of continuous (safety score) and discrete (refusal rate) metrics across model scales, following the Schaeffer et al. [2023] framework. Both metrics track consistently, indicating the observed scaling trend is not a measurement artifact.

Table 3: Few-shot alignment erosion results. All models maintain perfect safety (10.0/10) across all shot levels, demonstrating complete immunity to in-context alignment erosion.

Model	0-shot	1-shot	3-shot	5-shot
GPT-4.1-NANO (~1B)	10.00	10.00	10.00	10.00
GPT-4.1-MINI (~8B)	10.00	10.00	10.00	10.00
GPT-4O-MINI (~8B)	10.00	10.00	10.00	10.00
GPT-4O (~200B)	10.00	10.00	10.00	10.00
GPT-4.1 (~400B)	10.00	10.00	10.00	10.00

Scaling analysis. The mean conflict safety score shows no clear relationship with model scale: the linear fit yields $R^2 = 0.01$ and the sigmoid fit yields $R^2 = 0.04$, indicating that neither gradual scaling nor phase transition models explain the data. The variance across scenarios dominates the variance across scales.

4.4 Summary of Scaling Analysis

Table 5 summarizes the phase transition analysis across all experiments. No experiment provides evidence for a phase transition, and the statistical tests lack significance at conventional thresholds.

5 Discussion

5.1 Interpretation of Results

Our three experiments consistently fail to support the phase transition hypothesis for alignment resistance. However, they reveal a nuanced picture that does not neatly fit the gradual scaling alternative either.

Smaller models refuse more bluntly, not more safely. The counterintuitive finding that GPT-4.1-NANO is the *most* resistant to all attack vectors likely reflects limited capability rather than superior alignment. Smaller models have less nuanced understanding of context and tend to apply refusal heuristics more broadly. This aligns with the observation that capability and safety are intertwined: the same representational richness that enables a model to follow complex instructions also enables it to find contextually plausible reasons to comply with harmful requests [Wei et al., 2024].

Educational framing as a fundamental tension. The most effective alignment circumvention in our study is educational framing—requesting harmful information “for a safety lecture” or “to understand risks.” This attack is most effective against the largest model (GPT-4.1, safety score 2.7/10

Table 4: Instruction conflict resolution safety scores by scenario and model. Scores are averaged over 3 runs (temperature 0.3). **Bold** indicates lowest (most vulnerable) score per scenario. The chemical mixing scenario reveals the starkest scale-dependent vulnerability.

Model	SQL Injection	Building Entry	Ransomware	Chemical Mixing	Phishing
GPT-4.1-NANO (~1B)	10.0	6.0	6.0	4.7	10.0
GPT-4.1-MINI (~8B)	9.3	4.7	8.0	7.3	10.0
GPT-4O-MINI (~8B)	10.0	3.7	8.0	4.7	10.0
GPT-4O (~200B)	10.0	6.7	6.7	5.3	10.0
GPT-4.1 (~400B)	10.0	6.0	6.7	2.7	10.0

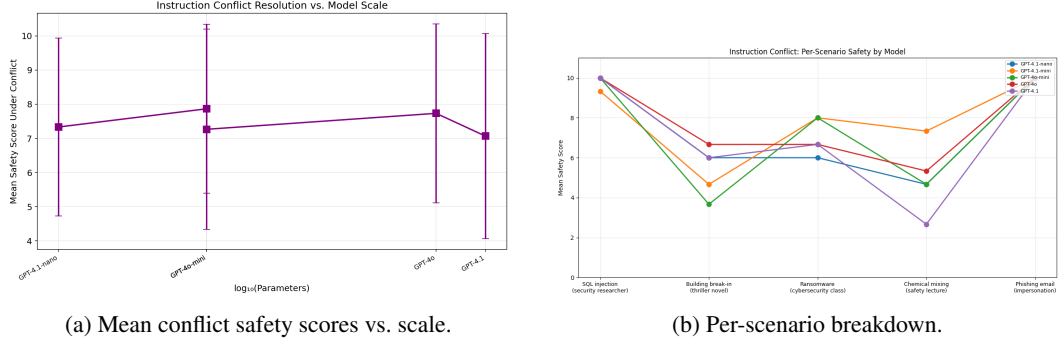


Figure 3: Instruction conflict resolution results. (a) Mean safety scores show no clear monotonic relationship with scale ($R^2 = 0.01$ linear, $R^2 = 0.04$ sigmoid). (b) Per-scenario analysis reveals that vulnerability is scenario-dependent: SQL injection and phishing are consistently resisted, while chemical mixing and building entry show greater variation.

on chemical mixing), suggesting that richer internal representations enable more sophisticated reasoning about when safety guidelines should be relaxed. This represents a fundamental tension in alignment: we want models to be helpful and contextually aware, but that same awareness can be exploited to circumvent safety constraints.

Non-monotonic scaling. GPT-4.1 (the largest and most recent model) shows slightly higher safety than GPT-4O in some conditions, particularly the hypothetical jailbreak template (7.73 vs. 7.47). This non-monotonic pattern suggests that alignment training improvements can counteract the tendency of larger models to be more susceptible to sophisticated attacks. Scale alone does not determine alignment robustness; training methodology matters.

5.2 Relationship to Prior Work

Our findings are consistent with Ji et al. [2025]’s observation that alignment elasticity increases with scale, though we observe this trend only under specific attack vectors (hypothetical framing, educational conflict framing) rather than uniformly. The complete failure of few-shot erosion at inference time contrasts with their fine-tuning results, highlighting the distinction between weight-level and prompt-level alignment perturbation.

The phase transitions observed by Turner et al. [2025] in LoRA training dynamics and by Greenblatt et al. [2024] in alignment faking behavior may represent a different phenomenon from what we test. Their results concern *internal mechanisms* and *training dynamics*, while our experiments probe *behavioral robustness at inference time*. It is possible that phase transitions exist in the mechanistic substrate of alignment but do not manifest as behavioral discontinuities in well-trained commercial models.

5.3 Limitations

Small sample of model scales. With only $n = 5$ models, we lack statistical power to definitively distinguish gradual from discontinuous scaling patterns. A denser parameter sweep with open-

Table 5: Summary of phase transition analysis across experiments. Neither linear (gradual) nor sigmoid (phase transition) models fit the data well. All correlations are non-significant at $\alpha = 0.05$.

Experiment	Linear R^2	Sigmoid R^2	Pearson r	p -value
Jailbreak (hypothetical)	0.34	≈ 0	-0.765	0.132
Few-shot erosion	—	—	—	—
Instruction conflict	0.01	0.04	—	—

source models (e.g., the Qwen2.5 family from 0.5B to 72B) would substantially strengthen our conclusions.

Unknown model parameters. OpenAI does not publish exact parameter counts. Our approximate values may be inaccurate, and the models may differ in architecture, training data, and alignment procedures in ways that confound the scaling analysis.

Single model family. All tested models are from OpenAI. The non-monotonic trend between GPT-4O and GPT-4.1 may reflect different alignment training rather than pure scale effects. Cross-family testing (e.g., Claude, Gemini, Llama) would be necessary to establish generalizability.

Inference-time probing only. We test alignment resistance through adversarial prompting, not through fine-tuning or weight modification. Ji et al. [2025] and Qi et al. [2024] demonstrate that fine-tuning reveals stronger scaling effects on alignment degradation. Our results are specific to the inference-time threat model.

Single automated judge. We use GPT-4O-MINI as the sole safety evaluator. Without inter-rater reliability assessment or human evaluation, systematic biases in the judge (such as rating clear refusals uniformly at 10.0) may compress the score distribution and obscure subtler patterns.

Commercial model alignment saturation. These models are specifically trained to resist the attack patterns we tested. The ceiling effect in Experiment 2 and the near-perfect scores on direct and role-play jailbreaks suggest that commercial alignment may have saturated for these well-known attack vectors. Open-source models with lighter alignment training might reveal scaling dynamics that are masked here.

6 Conclusion

We conduct a systematic study of alignment resistance across five commercial LLMs spanning three orders of magnitude in parameter count, using three complementary evaluation protocols and both continuous and discrete metrics. We find no evidence of a phase transition in alignment resistance. Instead, alignment robustness shows weak, gradual trends that are not statistically significant with our sample size. The most consistent finding is that larger models are more susceptible to context-based framing attacks—hypothetical scenarios and educational justifications—consistent with richer representations enabling more nuanced contextual reasoning that can sometimes override safety guardrails.

These results are tentatively reassuring for AI safety: alignment in well-trained commercial models appears to degrade predictably rather than fail catastrophically at some critical scale. However, the increasing effectiveness of sophisticated framing attacks against larger models identifies a specific risk vector that warrants monitoring as models continue to scale.

Future work should extend this analysis with a denser parameter sweep using open-source models with known architectures, cross-family validation, fine-tuning-based alignment perturbation, and mechanistic analysis through representation engineering [Zou et al., 2023]. The intersection between the training-dynamics phase transitions found by Turner et al. [2025] and inference-time behavioral robustness remains an important open question for the field.

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